MIL-E-1/796B
8 July 1977
SUPERSEDING
MIL-E-1/796A(NAVY)
12 October 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6024

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: ATR, Waveguide Type, F= 2800 MHz, peak incident power 1.0 kw

ABSOLUTE RATINGS:

Parameter:	Transmitter po	TA	Alt
Unit:	kw	°C	ft
Maximum:	1,000	+100	10,000
Minimum:	20	-40	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting position:	Any - See note 2

TEST CONDITIONS:

Parameter:	Transmitter po	tp	Du	VSWR
Unit:	kw	µs	---	----
Test 1:	20	1.0 ±10%	0.001	1.05
Test 2:	750	1.0 ±10%	0.001	1.20

Test Frequency: F = 2800 ±5.0%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
4021	Degradation due to vibration	--		---	---	---	---
4461	Loaded Q	--	Note 2	QL	---	5.5	---
	Quality conformance inspection, part 1		Note 4				
4482	Tuning susceptance	--	Note 2	b	-0.05	+0.05	---
4486	Firing time	1	Notes 2 and 3	t	---	10	sec
4488	Arc loss	1	Notes 2 and 3	La	---	0.8	dB
1027	Temperature cycling (nonoperating)	--	Note 1	---	---	---	---
1027	Temperature cycling life test	--	Group C, Note 1	---	50	---	Hz
	Quality conformance inspection, part 2						
4474	High-level VSWR	1		---	---	1.15	---
4484	Normalized conductance	--	Note 2	g	---	0.05	---
	Quality conformance inspection, part 3						
4551	Life test	2	Group D, Note 2	t	1,000	---	hrs
----	Life-test end points:	--					
4482	Tuning susceptance	--	Note 2	b	-0.05	+0.05	---
4484	Normalized conductance	--	Note 2	g	---	0.1	---
4486	Firing time	1	Notes 2 and 3	t	---	10	sec

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NOTES:

1. Satisfactory test on any one of 1B44, 1B52, 1B53, 1B56 1B57, 5921 or 5922 ATR tube types made during the same two week period, shall be evidence that all of the group meet this test requirement.

2. Mount in accordance with Drawing 153-JAN.

3. The tube shall be mounted as in note 2 and followed by a matched load. The tube shall fire within the limits specified after application of rf power. This test shall be performed at least 7 days after pumping and at least 24 hours after any previous discharge.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Review activities:

Army - MU
Air Force - 99

User activities:

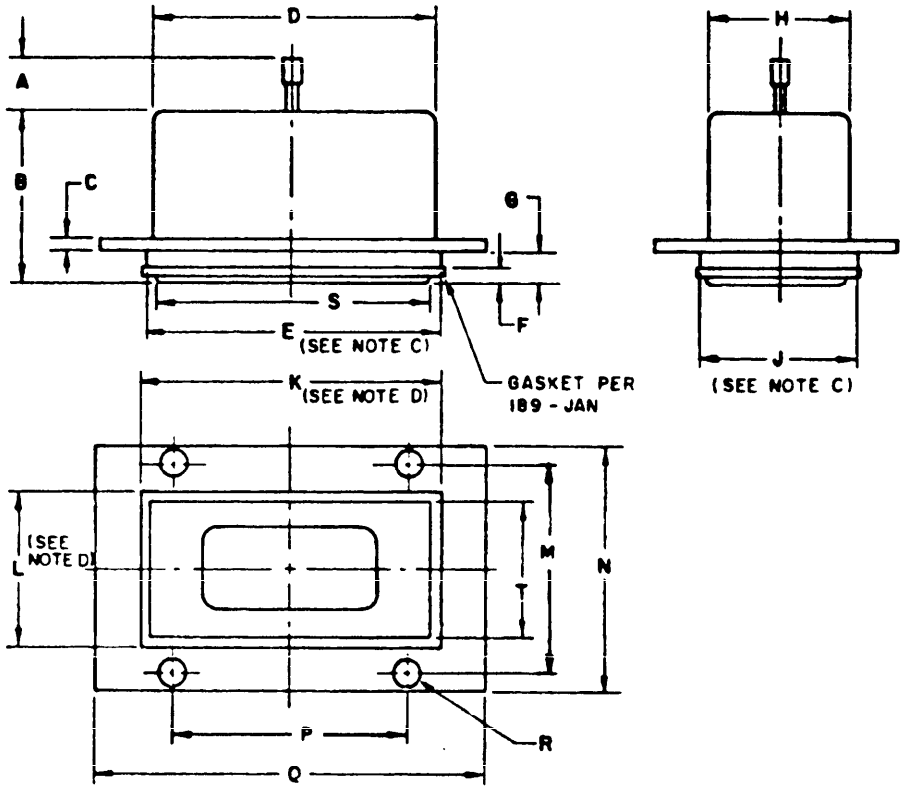
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Preparing activity:

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NOTES:

A Dimensions are in inches.

B Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

C Dimensions "E" and "J" pertain to flange.

D Dimensions "K" and "L" pertain to outside of gasket.

E Dimension "F" shall be measured prior to attachment of the gasket.

F Gasket to be securely mounted to tube.

FIGURE 1. Outline Drawing for Tube Type 6024.

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LTR	DIMENSIONS			
	INCHES		MILLIMETERS	
	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
Quality conformance inspection, part 1				
A		.625		15.88
B		1.875		47.62
C	.109	.141	2.77	3.58
D	2.985	3.015	75.82	76.58
F	.165 See Note E	.195 See Note E	4.19	4.95
G	.347	.377	8.81	9.58
H	1.486	1.516	37.74	38.51
K		3.223		81.86
L		1.725		43.82
N	2.609	2.641	66.27	67.08
Q	4.109	4.141	104.37	105.18
R	.203 (4 holes)	.245 DIA. (4 holes)	5.16	6.22
S	2.990	3.010	75.95	76.45
T	1.490	1.510	37.85	38.35
Quality conformance inspection, part 2				
E	3.203	3.233	81.36	82.12
J	1.703	1.733	43.26	44.02
M	2.15	2.35	50.60	59.69
P	2.40	2.60	61.06	66.04

FIGURE 1. Outline Drawing for Tube Type 6024. -Continued

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MIL-E-1/796B Electron Tube, Gas Switching Type 6024 FSC 5960

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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